

Replayable Memory and the Cognitive Basis of Abstraction

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Abstract

Abstraction is often treated as a linguistic or symbolic capacity, emerging late in cognitive development and absent in non-human animals. At the same time, artificial systems demonstrate strong compression and generalization without exhibiting robust abstraction. This paper argues that abstraction is neither linguistic nor representational in origin, but arises from a more basic cognitive capacity: replayable memory of construction. We propose that abstraction consists in stabilizing semantic equivalence across distinct experiential histories, and that this stabilization requires access to replayable internal traces. This account explains why abstraction appears early in development, is present in animals, and remains elusive in many artificial systems despite scale and data. The framework integrates findings from memory research, learning theory, and comparative cognition, offering a unifying account of abstraction as a functional, pre-linguistic capacity.

1 Introduction

Abstraction is central to human cognition. It allows individuals to recognize sameness across variation, to generalize beyond surface features, and to transfer knowledge across contexts. Traditionally, abstraction has been associated with symbolic reasoning, language, or explicit concept formation. On this view, abstraction is a late-emerging cognitive achievement layered atop more basic perceptual and associative processes.

At the same time, empirical evidence challenges this picture. Infants exhibit abstraction-like capacities prior to language acquisition. Non-human animals demonstrate flexible generalization across perceptually dissimilar stimuli. Conversely, many artificial systems that excel at pattern recognition and statistical learning fail to exhibit robust abstraction under recombination or perturbation.

These observations suggest that abstraction is neither equivalent to language nor reducible to compression. This paper proposes an alternative account: abstraction arises from replayable memory of construction. On this view, abstraction is a functional property of systems that can reconstruct, compare, and stabilize the consequences of their own past interactions.

2 Compression and Its Cognitive Limits

Compression plays a central role in theories of learning and memory. From classical associative learning to modern neural network models, cognitive efficiency is often understood in terms of reducing redundancy and extracting regularities from experience.

Compression supports prediction and generalization within familiar regimes. However, compression alone does not determine which regularities should remain stable under novel conditions. A compressed representation may entangle causal and incidental features, yielding brittle behavior when circumstances change.

In biological cognition, compression is constrained by additional mechanisms that regulate which distinctions matter. These mechanisms are not purely statistical. They depend on how experiences were generated, not merely on their aggregate structure.

This distinction becomes evident in developmental trajectories. Children often overgeneralize early, then refine abstractions through interaction and feedback. What changes is not merely representational efficiency, but sensitivity to the consequences of alternative actions and interpretations.

3 Abstraction as Semantic Stability

We define abstraction in terms of semantic stability rather than representational form.

Definition 3.1 (Semantic Effect). *A semantic effect is any consequence of an internal structure on future perception, action, inference, or learning.*

Definition 3.2 (Semantic Equivalence). *Two internal structures are semantically equivalent if they produce indistinguishable semantic effects across the system’s future interactions.*

Abstraction consists in recognizing and enforcing such equivalence. This recognition reduces the number of distinct internal states that must be tracked, without collapsing distinctions that matter for future behavior.

Crucially, semantic equivalence is counterfactual. It concerns how the system would behave under alternative trajectories, not only what it has already experienced. Determining such equivalence requires access to the processes by which internal structures were formed.

4 Replayable Memory

Memory is often treated as storage: a repository of past states or experiences. However, extensive evidence from neuroscience and psychology suggests that memory is reconstructive rather than archival. Past experiences are replayed, reactivated, and recombined in ways that support learning, planning, and generalization.

We formalize this capacity as replayable memory.

Definition 4.1 (Replayable Memory). *Replayable memory is the capacity of a system to reconstruct the semantic consequences of past experiential histories, potentially under variation or recombination.*

Replay need not be exact or deterministic at the neural level. What matters is functional access to the structure of past interactions sufficient to evaluate alternative continuations.

Proposition 4.1. *A system lacking replayable memory cannot, in general, determine semantic equivalence across distinct experiential histories.*

Proof. Semantic equivalence depends on future consequences under alternative trajectories. Without replayable access to how internal structures were formed, the system lacks the information required to evaluate such counterfactuals. Equivalence judgments must therefore rely on surface similarity or statistical coincidence. $\square$

Replayable memory provides a mechanism by which abstraction can be grounded in experience rather than imposed symbolically.

5 Developmental and Comparative Evidence

The proposed account aligns with evidence from developmental psychology. Infants demonstrate abstraction-like abilities before acquiring language, such as recognizing numerical equivalence or object permanence across transformations. These capacities emerge alongside the maturation of memory systems that support replay and consolidation.

In comparative cognition, animals exhibit abstraction in domains relevant to survival, such as tool use or social inference, despite lacking symbolic language. These abilities are closely tied to experience-dependent learning and memory replay, particularly during rest or sleep.

The present framework predicts that abstraction should correlate with the capacity for replayable memory rather than with linguistic ability. This prediction is consistent with observed dissociations between language competence and abstract reasoning across species and developmental stages.

6 Implications for Artificial Systems

Artificial systems trained via large-scale optimization often exhibit impressive compression but limited abstraction. They generalize within the training distribution but struggle with compositional novelty or causal intervention.

From the present perspective, this limitation arises because such systems lack replayable access to their own construction histories. Training traces are typically discarded once parameters are updated. The system retains the result of learning but not the process by which it was achieved.

This design choice precludes principled evaluation of semantic equivalence. Without replay, the system cannot distinguish stable invariants from accidental correlations. Abstraction must therefore be approximated indirectly, often with limited success.

The framework suggests that incorporating replayable memory into artificial systems—whether through explicit event traces, structured episodic memory, or reconstructive learning dynamics—may be necessary for achieving robust abstraction.

7 Discussion

By grounding abstraction in replayable memory, this account reconciles several otherwise puzzling observations. It explains why abstraction appears early in development, why it is present in animals, and why it remains elusive in many artificial systems despite scale.

Abstraction is not a matter of possessing symbols or concepts, but of having access to one’s own experiential construction. Language amplifies abstraction by externalizing and sharing stabilized equivalences, but it is not their source.

This view shifts the focus of cognitive theory from representational format to memory dynamics. What matters is not what is stored, but what can be reconstructed and compared.

8 Conclusion

We have proposed that abstraction is a functional consequence of replayable memory. By defining abstraction in terms of semantic equivalence across experiential histories, we obtain a framework that unifies findings from development, animal cognition, and artificial intelligence.

This account reframes abstraction as a pre-linguistic, experience-grounded capacity rather than a symbolic achievement. It also clarifies why compression alone is insufficient for abstraction, and why replayable construction plays a central role in intelligent cognition.

Understanding abstraction in these terms offers a principled basis for comparing natural and artificial systems and suggests concrete directions for future cognitive and computational research.